

Non-convex cone

Model — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.
- $\neg$ **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.

0.2 Construction

Let $X = \{a, b, c\}$ and identify a difference $q - p$ of lotteries with the point $(q(a) - p(a), q(b) - p(b)) \in \mathbb{R}^2$. Let $C \subseteq \mathbb{R}^2$ be the closed upper half-plane together with the closed sector of directions $[225^\circ, 270^\circ]$, and define $q \succcurlyeq p$ iff $q - p \in C$.

Completeness: $C \cup (-C) = \mathbb{R}^2$ since $-C$ contains the closed lower half-plane. Independence: mixing both sides with r multiplies the difference by $\lambda > 0$, and C is a cone. Closed graph: $(p, q) \mapsto q - p$ is continuous and C is closed. Reflexivity: $0 \in C$.

Transitivity fails: take x at direction 240° and y at direction 170° , both in C and small enough that $p, q = p + x, \dots$ are lotteries; $x + y$ has direction $\approx 205^\circ$, which is not in C . So with $q - p = x$ and $r - q = y$ we have $q \succcurlyeq p$ and $r \succcurlyeq q$ but not $r \succcurlyeq p$.

0.3 Notes

Sanity-checked numerically in `checks/countermodels.py` (completeness and independence hold, transitivity fails on random samples). Not yet checked by a human. Any non-convex closed cone C with $C \cup -C = \mathbb{R}^2$ works.

Record: `topics/decision-theory/models/non-convex-cone.yaml`